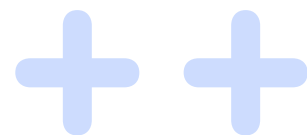


Engineering Devices Or Equipments In Health care



Transforming Healthcare,
Innovations and Challenges in
Modern Medicine



+ Introduction...

➤ **Engineering devices and equipment in healthcare are specialized electrical and electronic systems designed to support medical diagnosis, monitoring, and treatment through reliable, safe, and well-structured wiring infrastructure.**



➤ **Engineering advancements in the medical field improve accurate diagnosis, enhance treatment quality, ensure patient safety, enable advanced monitoring, and make healthcare faster, more reliable, and cost-effective.**



Types of Medical Devices...

➤ Diagnostic Devices

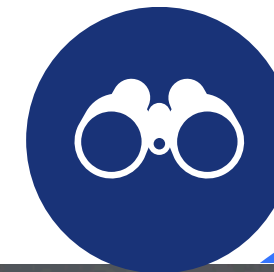
Used to detect diseases and analyze patient conditions.
Examples: MRI scanner, CT scan, X-ray machine, ECG sensors.

➤ Therapeutic Devices

Used for treatment or recovery support.
Examples: Pacemaker, infusion pump, defibrillator, dialysis machine.

➤ Monitoring Devices

Continuously track vital signs and patient health.
Examples: Ventilator, patient monitor, blood pressure monitor, glucose monitor.



Working Principle...



- ✓ **Devices work by converting biological signals into electrical/mechanical signals using engineering principles (pressure, light, magnetic field, motion).**
- ✓ **Use of biocompatible materials, electronic circuits, and sensors like ECG electrodes, pressure sensors, optical sensors.**
- ✓ **Software + microcontrollers process signals and control the device.**
- ✓ **Mechanical parts (motors, pumps, valves, transducers) help in movement, fluid flow, or energy conversion.**





Application And Impact On Health Care



Medical devices improve diagnosis, treatment, monitoring, and patient safety (e.g., MRI for accurate imaging, ventilators for life support, infusion pumps for precise drug delivery).

Lead to faster diagnosis, reduced human error, minimally invasive surgeries, and better patient outcomes.

Examples:

ECG → Detects heart abnormalities early

Robotic surgery → High precision, faster recovery

Pulse oximeter → Instant oxygen monitoring



References (APA/IEEE Format):

APA: World Health Organization. (2024). Medical devices: Global overview.

IEEE: [1] World Health Organization, “Medical Devices Overview,” 2024.





**Thank
You.**

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